TASK 01

LINEAR REGRESSION MODEL

A researcher wants to predict the life expectancy of countries based on socio-economic and health factors. You are tasked with building a Linear Regression model that can predict life expectancy using historical data

CODE AND OUTPUT :

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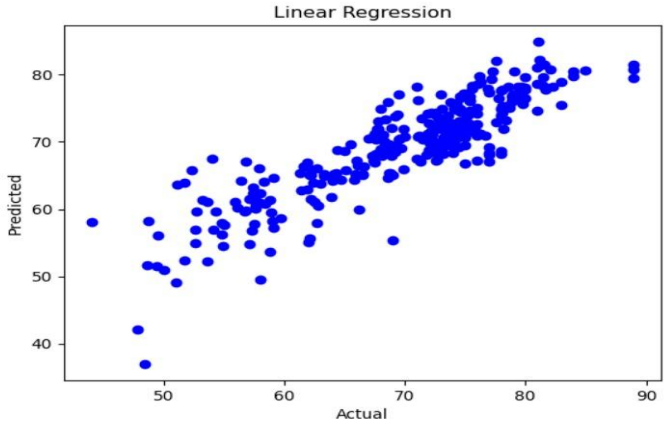
Code JupyterLab Python [conda env:base] * Anaconda Toolbox

[7]:
```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.linear_model import LinearRegression
from sklearn.model_selection import train_test_split
from sklearn.metrics import mean_squared_error, r2_score

# Load Data
df = pd.read_csv("Life Expectancy Data.csv")

# Remove missing values (simple method)
df = df.dropna()

# Prepare Data (use exact column names with spaces)
X = df[['Adult Mortality', 'Alcohol', 'GDP', 'Schooling', ' HIV/AIDS']]
y = df['Life expectancy ']

# Split
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)

# Model
model = LinearRegression()
model.fit(X_train, y_train)

# Predict
y_pred = model.predict(X_test)

# Visualization
plt.scatter(y_test, y_pred, color='blue')
plt.xlabel('Actual')
plt.ylabel('Predicted')
plt.title('Linear Regression')
plt.show()
```

```python
# Output
print(df.head())
print("MSE:", mean_squared_error(y_test, y_pred))
print("R2:", r2_score(y_test, y_pred))
print("Coefficient:", model.coef_)
print("Intercept:", model.intercept_)
```



Actual

```
        Country  Year      Status  Life expectancy   Adult Mortality  \
 0  Afghanistan  2015  Developing              65.0            263.0
 1  Afghanistan  2014  Developing              59.9            271.0
 2  Afghanistan  2013  Developing              59.9            268.0
 3  Afghanistan  2012  Developing              59.5            272.0
 4  Afghanistan  2011  Developing              59.2            275.0

    infant deaths  Alcohol  percentage expenditure  Hepatitis B  Measles   ...  \
 0             62     0.01               71.279624         65.0     1154  ...
 1             64     0.01               73.523582         62.0      492  ...
 2             66     0.01               73.219243         64.0      430  ...
 3             69     0.01               78.184215         67.0     2787  ...
 4             71     0.01                7.097109         68.0     3013  ...

    Polio  Total expenditure  Diphtheria    HIV/AIDS         GDP  Population  \
 0    6.0               8.16        65.0        0.1  584.259210  33736494.0
 1   58.0               8.18        62.0        0.1  612.696514    327582.0
 2   62.0               8.13        64.0        0.1  631.744976  31731688.0
 3   67.0               8.52        67.0        0.1  669.959000   3696958.0
 4   68.0               7.87        68.0        0.1   63.537231   2978599.0


4   68.0               7.87         68.0        0.1   63.537231   2978599.0

    thinness  1-19 years   thinness 5-9 years  \
0                  17.2                 17.3
1                  17.5                 17.5
2                  17.7                 17.7
3                  17.9                 18.0
4                  18.2                 18.2

   Income composition of resources  Schooling
0                            0.479       10.1
1                            0.476       10.0
2                            0.470        9.9
3                            0.463        9.8
4                            0.454        9.5

[5 rows x 22 columns]
MSE: 16.707257679098234
R2: 0.7647610077126137
Coefficient: [-2.02489404e-02 -4.81139617e-02  7.60170284e-05  1.59778507e+00
 -4.50679171e-01]
Intercept: 53.95235193835487
```